

831.

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**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

**Annexure A
SHORT ANSWER TEST**

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I K. Yoganand (192525173) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

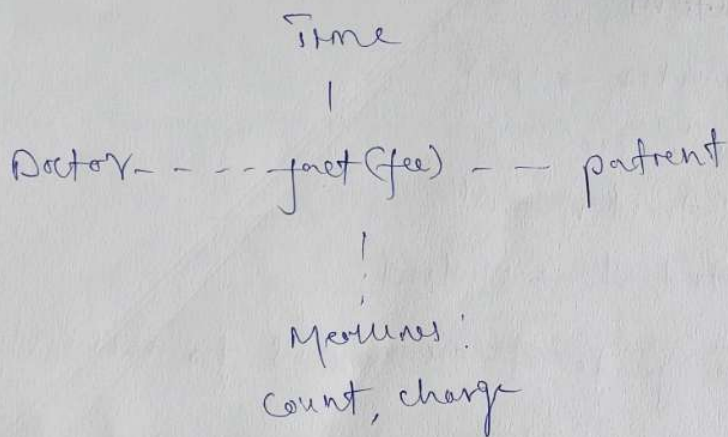
K. Yoganand
Signature of the student:

1 Data warehouse (Time, Doctor, patient)

(a) three schema types:-

- * star schema
- * snowflake schema
- * fact constellation (galaxy) schema

(b) star schema



(c) SQL Query

SELECT day, month, year, doctor, hospital, patient

SUM (count) AS total - count,

SUM (charge) AS total - charge

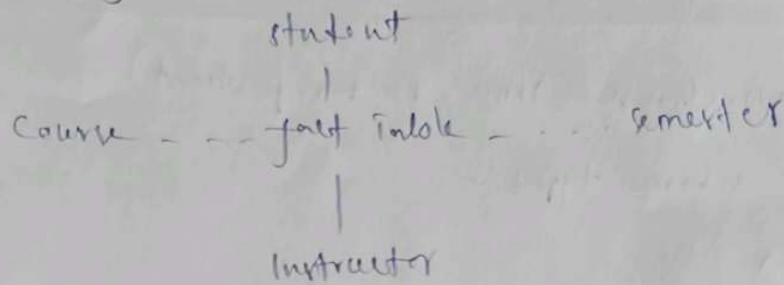
FROM fee

GROUP BY day, month, year, doctor,

hospital, patient;

2 Bg university data warehouse

(a) Snowflake schema



operations;

* Count

* Avg - Grade

(b) OLAP Operations

1. Roll-up semester $\rightarrow$ year
2. slice Course = 'csc'
3. Roll-up instructor $\rightarrow$ all
4. Roll up - student $\rightarrow$ student

Result:- Average grade of CSC Courses for each student

(c) Number of cuboids

Each dimension has 4 levels

formula;

number of cuboids

$$= 4^4 = 256$$

dimensions!

- * Time (hour $\rightarrow$ day $\rightarrow$ month $\rightarrow$ year)
- * Location (probe - city - Region)
- * weather type

measures!

- * Air pressure
- * Temperature
- * precipitation

schemas! star schema

this design supports efficient OLAP Operations such as roll-up, drill down, slice and dice

4

Data cube (n dimensions, each with p values)

(a) maximum base cuboid cells

$$p^n$$

(b) minimum base cuboid cells

0

(if no data exists)

(c) maximum cells in data cube

$$(p+1)^n$$

(d) minimum cells in the data cube

1

(the apex cuboid)

4

(a) the base cuboid is the most detailed level of the data cube, every combination of the value in all dimensions creates one cell

* number of dimensions = n

* Distinct value in each dimension = p

$$p^n \quad \text{ex: } n=3, p=4 = 4^3 = 64$$

(b) the maximum occurs when no data records are present in the cube

$\therefore$ if the database is empty, there are one cell

(c) A complete data cube includes

* the base cuboid

* all aggregate cuboids

* the apex cuboid (all dimensions)

$$\therefore (p+1)^n \quad \text{ex: } n=3, p=4 = (4+1)^3 = 5^3 = 125$$

(d) the smallest possible data cube contains only the apex cuboid, which represents the aggregation of all dimensions.